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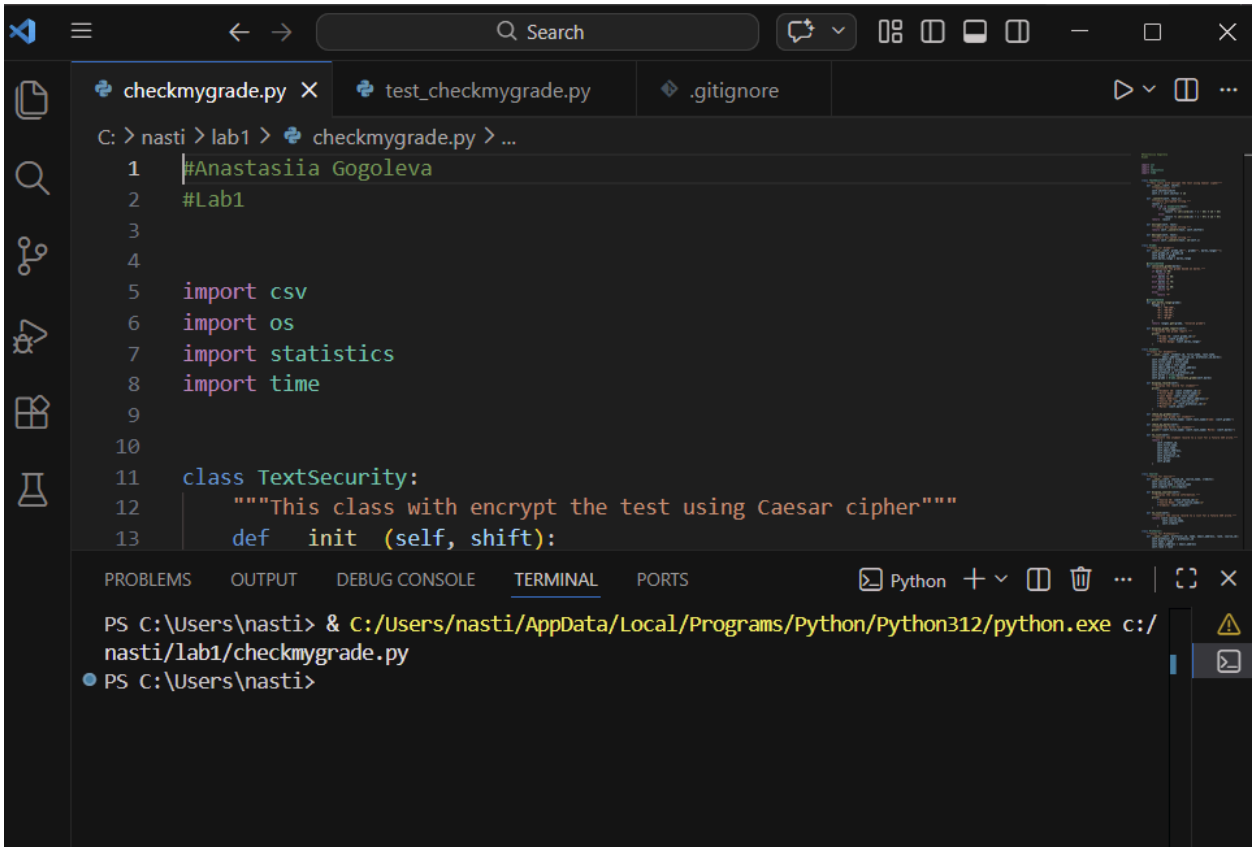
Lab 1

GitHub

https://github.com/AnanasSec/checkmygrade_app

Output after running the main file program.
It doesn't show any errors and runs smoothly.



Output after the test program

checkmygrade.py | test_checkmygrade.py

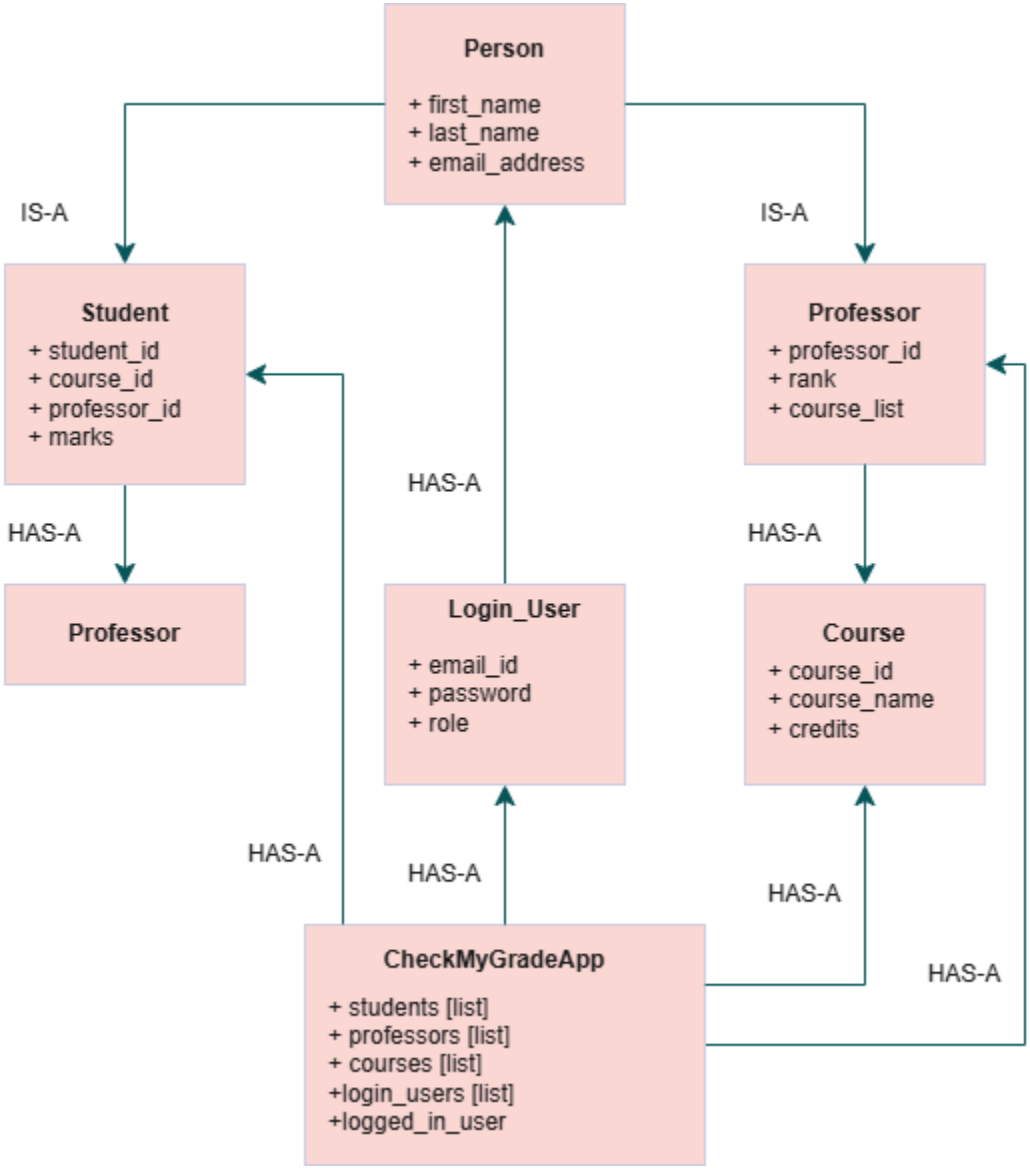
test_checkmygrade.py > TestCheckMyGradeApp > test_add_professor

```python
import unittest
from checkmygrade import CheckMyGradeApp


class TestCheckMyGradeApp(unittest.TestCase):

    def setUp(self):
        self.app = CheckMyGradeApp()

        self.app.students = []
        self.app.courses = []
        self.app.professors = []
        self.app.login users = []
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
    self.assertEqual(len(self.app.professors), 1) …
PS C:\Users\nasti\OneDrive\Desktop\DATA 200 python\lab1> & C:/Users/nasti/AppData/Local/Programs/Python
/Python312/python.exe "c:/Users/nasti/OneDrive/Desktop/DATA 200 python/lab1/test_checkmygrade.py"
......Search time for loaded CSV data: 0.00036100 seconds
...Sort by email time: 0.00021440 seconds
.Sort by marks time: 0.00006130 seconds
...
----------------------------------------------------------------------
Ran 13 tests in 4.922s

OK
PS C:\Users\nasti\OneDrive\Desktop\DATA 200 python\lab1>
```

3. Object Oriented Design (OOD) for CheckMyGrade Application



P.S.
I did some screenshots of the tests for functionalities inside the main file and I will attach the screenshots showing the functions functionalities.

```
PS C:\Users\nasti\OneDrive\Desktop\DATA 200 python\lab1> & C:/Users/nasti/AppData/Local/Programs/Python/Pyt
on312/python.exe "c:/Users/nasti/OneDrive/Desktop/DATA 200 python/lab1/checkmygrade.py"
Course added successfully.
Course added successfully.
Professor added successfully.
Professor added successfully.
Student added successfully.
Student added successfully.
Student added successfully.
Login user added successfully.
Login user added successfully.
Login successful for mcgonagal@hogwarts.edu.

All Student Records
Student ID: S101
First Name: Harry
Last Name: Potter
Email Address: potter@hogwarts.edu
Course ID: DATA 200
Professor ID: P101
Marks: 85.0
Student ID: S102
First Name: Hermione
Last Name: Granger
Email Address: granger@hogwarts.edu
Course ID: DATA 201
Professor ID: P102
Marks: 100.0
Student ID: S103
First Name: Ron
Last Name: Weasley
Email Address: ron@hogwarts.edu
Course ID: DATA 200
Professor ID: P101
Marks: 76.0
```

```
All Courses
Course ID: DATA 200
Course Name: Python Programming
Credits: 3
Course ID: DATA 201
Course Name: Databases
Credits: 3

All Professors
Professor ID: P101
Name: Minerva McGonagall
Email Address: mcgonagal@hogwarts.edu
Rank: Senior Professor
Course ID: DATA 200
Professor ID: P102
Name: Severus Snape
Email Address: snape@hogwarts.edu
Rank: Associate Professor
Course ID: DATA 201

Search for 'Hermione'
Search completed in 0.00001710 seconds
Student ID: S102
First Name: Hermione
Last Name: Granger
Email Address: granger@hogwarts.edu
Course ID: DATA 201
Professor ID: P102
Marks: 100.0
```

```
Sort Students by Name
Students sorted by name successfully.
Student ID: S102
First Name: Hermione
Last Name: Granger
Email Address: granger@hogwarts.edu
Course ID: DATA 201
Professor ID: P102
Marks: 100.0
Student ID: S101
First Name: Harry
Last Name: Potter
Email Address: potter@hogwarts.edu
Course ID: DATA 200
Professor ID: P101
Marks: 85.0
Student ID: S103
First Name: Ron
Last Name: Weasley
Email Address: ron@hogwarts.edu
Course ID: DATA 200
Professor ID: P101
Marks: 76.0
```

```
Sort Students by Marks
Students sorted by marks successfully.
Student ID: S102
First Name: Hermione
Last Name: Granger
Email Address: granger@hogwarts.edu
Course ID: DATA 201
Professor ID: P102
Marks: 100.0
Student ID: S101
First Name: Harry
Last Name: Potter
Email Address: potter@hogwarts.edu
Course ID: DATA 200
Professor ID: P101
Marks: 85.0
Student ID: S103
First Name: Ron
Last Name: Weasley
Email Address: ron@hogwarts.edu
Course ID: DATA 200
Professor ID: P101
Marks: 76.0
```

```
Update Student S103
Student updated successfully.
Report for Student ID: S103
Student ID: S103
First Name: Ron
Last Name: Weasley
Email Address: ron@hogwarts.edu
Course ID: DATA 200
Professor ID: P101
Marks: 88.0
```

```
Statistics
Average marks for course DATA 200: 86.50
Median marks for course DATA 200: 86.50
Report for Course ID: DATA 200
Student ID: S101
First Name: Harry
Last Name: Potter
Email Address: potter@hogwarts.edu
Course ID: DATA 200
Professor ID: P101
Marks: 85.0
Student ID: S103
First Name: Ron
Last Name: Weasley
Email Address: ron@hogwarts.edu
Course ID: DATA 200
Professor ID: P101
Marks: 88.0
Report for Professor ID: P101
Student ID: S101
First Name: Harry
Last Name: Potter
Email Address: potter@hogwarts.edu
Course ID: DATA 200
Professor ID: P101
Marks: 85.0
Student ID: S103
```

```
First Name: Ron
Last Name: Weasley
Email Address: ron@hogwarts.edu
Course ID: DATA 200
Professor ID: P101
Marks: 88.0
Report for Student ID: S101
Student ID: S101
First Name: Harry
Last Name: Potter
Email Address: potter@hogwarts.edu
Course ID: DATA 200
Professor ID: P101
Marks: 85.0

Change Password
Password changed successfully.
Login successful for potter@hogwarts.edu.
Logout successful.
PS C:\Users\nasti\OneDrive\Desktop\DATA 200 python\lab1>
```